

```
In [3]: import pandas as pd
import pickle

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_mat
```

```
In [4]: data = pd.read_csv(r"C:\Users\HP-PC\Desktop\Handling Outliers\dataset\Crop_recom
```

```
In [5]: X = data.drop("label", axis=1)
y = data["label"]
```

```
In [6]: X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
In [7]: model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)
```

```
Out[7]: ▼ RandomForestClassifier ⓘ ?
    ► Parameters
    ► Fitted attributes
```

```
In [8]: y_pred = model.predict(X_test)
```

```
In [9]: accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

Accuracy: 0.9931818181818182

```
In [10]: cr = classification_report(y_test, y_pred)
print(cr)
```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| apple        | 1.00      | 1.00   | 1.00     | 23      |
| banana       | 1.00      | 1.00   | 1.00     | 21      |
| blackgram    | 1.00      | 1.00   | 1.00     | 20      |
| chickpea     | 1.00      | 1.00   | 1.00     | 26      |
| coconut      | 1.00      | 1.00   | 1.00     | 27      |
| coffee       | 1.00      | 1.00   | 1.00     | 17      |
| cotton       | 1.00      | 1.00   | 1.00     | 17      |
| grapes       | 1.00      | 1.00   | 1.00     | 14      |
| jute         | 0.92      | 1.00   | 0.96     | 23      |
| kidneybeans  | 1.00      | 1.00   | 1.00     | 20      |
| lentil       | 0.92      | 1.00   | 0.96     | 11      |
| maize        | 1.00      | 1.00   | 1.00     | 21      |
| mango        | 1.00      | 1.00   | 1.00     | 19      |
| mothbeans    | 1.00      | 0.96   | 0.98     | 24      |
| mungbean     | 1.00      | 1.00   | 1.00     | 19      |
| muskmelon    | 1.00      | 1.00   | 1.00     | 17      |
| orange       | 1.00      | 1.00   | 1.00     | 14      |
| papaya       | 1.00      | 1.00   | 1.00     | 23      |
| pigeonpeas   | 1.00      | 1.00   | 1.00     | 23      |
| pomegranate  | 1.00      | 1.00   | 1.00     | 23      |
| rice         | 1.00      | 0.89   | 0.94     | 19      |
| watermelon   | 1.00      | 1.00   | 1.00     | 19      |
| accuracy     |           |        | 0.99     | 440     |
| macro avg    | 0.99      | 0.99   | 0.99     | 440     |
| weighted avg | 0.99      | 0.99   | 0.99     | 440     |

```
In [11]: cm = confusion_matrix(y_test, y_pred)
print(cm)
```

```
[[23  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0]
 [ 0 21  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0]
 [ 0  0 20  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0]
 [ 0  0  0 26  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0]
 [ 0  0  0  0 27  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0]
 [ 0  0  0  0  0 17  0  0  0  0  0  0  0  0  0  0  0  0  0  0]
 [ 0  0  0  0  0 17  0  0  0  0  0  0  0  0  0  0  0  0  0  0]
 [ 0  0  0  0  0  0 14  0  0  0  0  0  0  0  0  0  0  0  0  0]
 [ 0  0  0  0  0  0  0 23  0  0  0  0  0  0  0  0  0  0  0  0]
 [ 0  0  0  0  0  0  0  0 20  0  0  0  0  0  0  0  0  0  0  0]
 [ 0  0  0  0  0  0  0  0  0 11  0  0  0  0  0  0  0  0  0  0]
 [ 0  0  0  0  0  0  0  0  0  0 21  0  0  0  0  0  0  0  0  0]
 [ 0  0  0  0  0  0  0  0  0  0  0 19  0  0  0  0  0  0  0  0]
 [ 0  0  0  0  0  0  0  0  0  0  0  1 23  0  0  0  0  0  0  0]
 [ 0  0  0  0  0  0  0  0  0  0  0  0  0 19  0  0  0  0  0  0]
 [ 0  0  0  0  0  0  0  0  0  0  0  0  0  0 17  0  0  0  0  0]
 [ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 14  0  0  0  0]
 [ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 23  0  0  0]
 [ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 23  0  0]
 [ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 23  0]
 [ 0  0  0  0  0  0  0  0  2  0  0  0  0  0  0  0  0  0  0 17]
 [ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 19]]
```

```
In [12]: with open("model.pkl", "wb") as f:
pickle.dump(model, f)

print("Model saved successfully!")
```

Model saved successfully!

In [ ]: